

Marginal Distributions

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Introduction

Given a joint distribution of several random variables, the marginal distribution of any one of them is obtained by integrating (or summing) out the others. Marginalisation is the probability-theoretic version of “don’t care”: collapsing multi-dimensional information into a single dimension.

Prerequisites

Joint distributions, integration/summation over probability densities.

Theory

For a joint density $f_{X,Y}(x, y)$:

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy.$$

Analogously, for a joint PMF: $p_X(x) = \sum_y p_{X,Y}(x, y)$.

The joint CDF gives marginals by pushing the other argument to infinity: $F_X(x) = F_{X,Y}(x, \infty)$.

Key property: the marginals do not determine the joint. Many joint distributions share the same marginals but differ in dependence structure (verified by Sklar’s theorem: marginals + copula = joint).

For a multivariate normal, the marginal of each component is univariate normal with the corresponding mean and variance; more generally, any subset of a multivariate normal is multivariate normal.

Assumptions

Integrability of the joint density; otherwise standard probability-space assumptions.

R Implementation

```
library(mvtnorm)
set.seed(2026)

# Bivariate normal with correlation 0.6
Sigma <- matrix(c(1, 0.6, 0.6, 1), 2, 2)
XY <- rmvnorm(5000, sigma = Sigma)

# Marginal distributions of X and Y
c(mean_X = mean(XY[, 1]), sd_X = sd(XY[, 1]),
  mean_Y = mean(XY[, 2]), sd_Y = sd(XY[, 2]))

# Each marginal is standard normal regardless of correlation
ks.test(XY[, 1], "pnorm")
```

```
ks.test(XY[, 2], "pnorm")

# Demonstrate: same marginals, different joints
# Joint 1: correlation 0.6 (bivariate normal)
# Joint 2: same marginals but comonotonic (perfect rank correlation)
U <- pnorm(XY[, 1]) # uniform marginal via probability integral transform
X1 <- qnorm(U); Y1 <- qnorm(U) # perfectly dependent copy
cov(XY)
cov(cbind(X1, Y1))
```

Output & Results

mean_X	sd_X	mean_Y	sd_Y
0.0099	1.0050	-0.0027	1.0018

```
One-sample Kolmogorov-Smirnov test
D = 0.012, p-value = 0.4672 # X marginal matches N(0,1)
```

```
One-sample Kolmogorov-Smirnov test
D = 0.015, p-value = 0.2132 # Y marginal matches N(0,1)
```

	[,1]	[,2]
[1,]	1.00	0.61
[2,]	0.61	1.00

	X1	Y1	
X1	1.00	1.00	# comonotonic: correlation 1 with same marginals
Y1	1.00	1.00	

Same marginals (both standard normal), very different joint distributions: correlation 0.6 versus perfect dependence.

Interpretation

In reporting, marginal summaries (marginal mean, SD) are usually what Table 1 presents. When dependence matters – pair-of-eyes measurements, twin pairs, repeated measures – a marginal analysis is incomplete and must be complemented with joint or conditional reporting.

Practical Tips

- Always report marginals alongside any joint analysis; they provide basic sanity checks.
- Marginalisation is trivial for multivariate normals: take the relevant subvector of the mean and the submatrix of the covariance.
- For continuous joint densities without closed form, marginalise numerically via `integrate()` or Monte Carlo.
- Empirical marginals are obtained by projection: for a sample $\{(X_i, Y_i)\}$, the marginal sample of X is $\{X_i\}$.
- In latent-variable models, marginalising over the latent gives the observed-data likelihood – the central quantity in EM and MCMC.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.

- Diagnostics that should be reported but often aren't.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- Probability, conditional probability, Bayes' theorem
- Discrete distributions: Bernoulli, binomial, Poisson
- Continuous distributions and Q-Q plots